

Contractively split compact projective condensed abelian groups are free

Candidate partial result for arXiv:2605.11731

9 August 2026

Status. Candidate partial result; likely valid; human review requested. This packet does not claim a full answer to the source question.

1 Source question

Clausen and Scholze prove that every compact projective condensed abelian group is a retract of $\mathbb{Z}[S]$ for some extremally disconnected profinite set S , and then ask the following question [1, Question 3.6, p. 26]:

Are all compact projective condensed abelian groups isomorphic to $\mathbb{Z}[S]$ for some $S \in \text{Extr}$?

to find an isomorphism (after reCompleting)

$$\mathbb{Z}[\beta(I \times \mathbb{N})] \cong M \oplus \mathbb{Z}[\beta(I \times \mathbb{N})]. \quad \square$$

The following question is open, and closely related to an open question on injective Banach spaces [\[ASC⁺16\]](#) Section 1.6.1].

QUESTION 3.6. Are all compact projective condensed abelian groups isomorphic to $\mathbb{Z}[S]$ for some $S \in \text{Extr}$?

We note that one can find non-extremally disconnected S such that $\mathbb{Z}[S]$ is compact projective; for example, if S is obtained by identifying two points in the boundary of $\beta\mathbb{N}$, then $\mathbb{Z}[S] \cong \mathbb{Z}[\beta\mathbb{N}]$.

In any case, this indicates that there are extremely few compact projectives. They are also destroyed by basic operations:

The source emphasizes that the unrestricted question is closely related to an old question on the structure of injective Banach spaces [2, Section 1.6.1].

2 Definitions and statement

Write a profinite set as $S = \varprojlim_i S_i$, with each S_i finite. The explicit description used in the source is

$$\mathbb{Z}[S] = \bigcup_{n \geq 0} \mathcal{B}_n(S), \quad \mathcal{B}_n(S) := \varprojlim_i \{a \in \mathbb{Z}[S_i] : \|a\|_1 \leq n\}.$$

An endomorphism $e : \mathbb{Z}[S] \rightarrow \mathbb{Z}[S]$ is determined, under the free/forgetful adjunction, by its *generator map*

$$g_e : S \longrightarrow \mathbb{Z}[S].$$

We call e *generator-contractive* if g_e factors through $\mathcal{B}_1(S)$. This is the integral radius-one condition in the canonical ℓ^1 filtration; it does not mean that every presentation of the same retract has this property.

Theorem 1 (Radius-one classification). *Let $S \in \text{Extr}$ and let $e : \mathbb{Z}[S] \rightarrow \mathbb{Z}[S]$ be a generator-contractive idempotent. There is an extremally disconnected profinite retract $T \subseteq S$ such that*

$$\text{im}(e) \cong \mathbb{Z}[T].$$

More precisely, there is a clopen partition $S = S_0 \sqcup S_+ \sqcup S_-$ and a continuous map $r : S_+ \sqcup S_- \rightarrow T$, with $T \subseteq S_+$ and $r|_T = \text{id}_T$, for which

$$g_e(s) = \begin{cases} 0, & s \in S_0, \\ [r(s)], & s \in S_+, \\ -[r(s)], & s \in S_-. \end{cases}$$

Conversely, every signed partial retraction of this form induces a generator-contractive idempotent with image $\mathbb{Z}[T]$.

Consequently, Question 3.6 has an affirmative answer for every compact projective which admits a presentation as the image of a generator-contractive idempotent on some $\mathbb{Z}[S]$.

3 Proof intuition

At integral ℓ^1 -radius one there are only three kinds of elements: zero, a positive basis vector, and a negative basis vector. Hence the generator map of e is a continuous signed partial map from S back to S . Idempotence is unexpectedly rigid: every point hit by this partial map must be hit positively and fixed. Those fixed points form a retract T of S . The signed partial map then linearizes to a retraction $\mathbb{Z}[S] \rightarrow \mathbb{Z}[T]$, and the inclusion $T \hookrightarrow S$ linearizes to its section.

This mechanism is special to radius one. Starting at radius two, a generator may map to a sum or difference of basis vectors, so there is no point-valued map whose fixed set can be extracted.

4 Proof

Lemma 1. *For every profinite set S , there is a natural isomorphism of profinite sets*

$$\mathcal{B}_1(S) \cong \{0\} \sqcup S^+ \sqcup S^-,$$

where the two copies of S send s to $[s]$ and $-[s]$ respectively.

Proof. For a finite set S_i , an integral vector of ℓ^1 -norm at most one is exactly 0, $[s_i]$, or $-[s_i]$. Under every transition map between finite quotients, zero remains zero and the sign and total coefficient of a nonzero vector remain fixed. A compatible family of positive (respectively negative) basis vectors is exactly a point of $S = \varprojlim_i S_i$. Taking the inverse limit gives the claimed natural decomposition. $\square$

Proof of Theorem 1. By Lemma 1, the continuous map $g_e : S \rightarrow \mathcal{B}_1(S)$ determines a clopen partition

$$S = S_0 \sqcup S_+ \sqcup S_-$$

and a continuous map $r : U := S_+ \sqcup S_- \rightarrow S$ such that $g_e(s) = 0, [r(s)], -[r(s)]$ on the respective pieces.

Evaluate $e^2 = e$ on the universal generator $[s]$. If $s \in S_+$ and $t = r(s)$, then

$$g_e(t) = e([t]) = e(g_e(s)) = g_e(s) = [t].$$

If $s \in S_-$ and $t = r(s)$, then

$$-g_e(t) = e(-[t]) = e(g_e(s)) = g_e(s) = -[t],$$

so again $g_e(t) = [t]$. Thus every point in $r(U)$ is positively fixed. Put

$$T := \{t \in S_+ : r(t) = t\}.$$

The preceding calculation gives $r(U) \subseteq T$, while the reverse inclusion follows from $r(t) = t$ for $t \in T$. Hence $r(U) = T$ and $r|_T = \text{id}_T$. The set T is closed, being an equalizer inside the clopen profinite set S_+ .

If T is empty, then U is empty and $e = 0$, giving $\text{im}(e) = \mathbb{Z}[\emptyset]$. Otherwise choose $t_0 \in T$. Extend r across the clopen piece S_0 by the constant value t_0 . This produces a continuous retraction $S \rightarrow T$. A retract of an extremally disconnected compact Hausdorff space is extremally disconnected: for an open $V \subseteq T$, the closure of V in T is the intersection with T of the closure of its inverse image under the retraction, and is therefore open. Thus $T \in \text{Extr}$.

Let $i : \mathbb{Z}[T] \rightarrow \mathbb{Z}[S]$ be induced by the inclusion. Define $p : \mathbb{Z}[S] \rightarrow \mathbb{Z}[T]$ from the generator map

$$s \mapsto \begin{cases} 0, & s \in S_0, \\ [r(s)], & s \in S_+, \\ -[r(s)], & s \in S_-. \end{cases}$$

This map is continuous by the clopen decomposition and Lemma 1 applied to T . On universal generators, $i \circ p$ agrees with e , while $p \circ i$ is the identity because every $t \in T$ lies in S_+ and is fixed by r . Hence $e = i \circ p$, $p \circ i = \text{id}_{\mathbb{Z}[T]}$, and the image of e is isomorphic to $\mathbb{Z}[T]$.

Conversely, the same formulas attached to any signed partial retraction give $p \circ i = \text{id}$, so $e = i \circ p$ is an idempotent, is generator-contractive, and has image $\mathbb{Z}[T]$. $\square$

5 Verification and stress tests

The argument is exact and uses no computation. As a finite sanity check, the script `code/check_finite_contractive_idempotents.py` enumerates every map from an n -point set to $\{0\} \sqcup S^+ \sqcup S^-$ for $1 \leq n \leq 5$, keeps the idempotents, and checks that every nonzero output targets a positively fixed point. It checked respectively

$$2, \quad 8, \quad 44, \quad 288, \quad 2192$$

idempotents. This is corroboration of the pointwise algebra only; it is not a proof of the continuity or extremal-disconnectedness steps.

The most useful hand checks are:

- Negative outputs cause no sign error: $e(-[t]) = -e([t])$, so idempotence still forces $e([t]) = [t]$.
- If $e \neq 0$, then $T \neq \emptyset$, because every nonzero output lands in T .
- The extension over S_0 is continuous because S_0 is clopen.
- Equality of the constructed morphisms can be checked on the universal generator by the free/forgetful adjunction.

6 Limitations and full-result frontier

Every compact projective is the image of some idempotent on $\mathbb{Z}[S]$, and the quasicompactness of S places its generator map in $\mathcal{B}_N(S)$ for some finite N . The theorem settles the sharp first layer $N = 1$.

The proof does not extend formally to $N = 2$. Already on two points, the idempotent

$$e([a]) = [a] + [b], \quad e([b]) = 0$$

has generator bound two. Its image is free in this finite example, but it has no positively fixed *basis point*; the fixed object is instead generated by the vector $[a] + [b]$. In a profinite family, the missing step is to choose such integral image bases continuously. That is exactly the new structural issue at higher radius, and the source’s comparison with complemented subspaces/injective Banach spaces cautions against treating it as routine.

A bounded literature search on 9 August 2026 used the exact wording of Question 3.6 and combinations of “compact projective condensed abelian group”, “idempotent”, “contractive”, “ ℓ^1 ”, and “retract”. It covered the source notes, recent condensed-mathematics notes, and the cited injective-Banach context. No matching radius-one classification was found; novelty is therefore provisional.

7 Human-review recommendation

The candidate is likely valid and merits review as a partial result. Review should concentrate on Lemma 1, the deduction $r(U) = T$, and the topological assertion that the constructed T is an extremally disconnected retract. The unrestricted Question 3.6 remains open.

References

- [1] D. Clausen and P. Scholze, *Condensed Mathematics and Complex Geometry*, arXiv:2605.11731, 2026.
- [2] A. Avilés, F. Cabello Sánchez, J. M. F. Castillo, M. González, and Y. Moreno, *Separably Injective Banach Spaces*, Lecture Notes in Mathematics 2132, Springer, 2016.